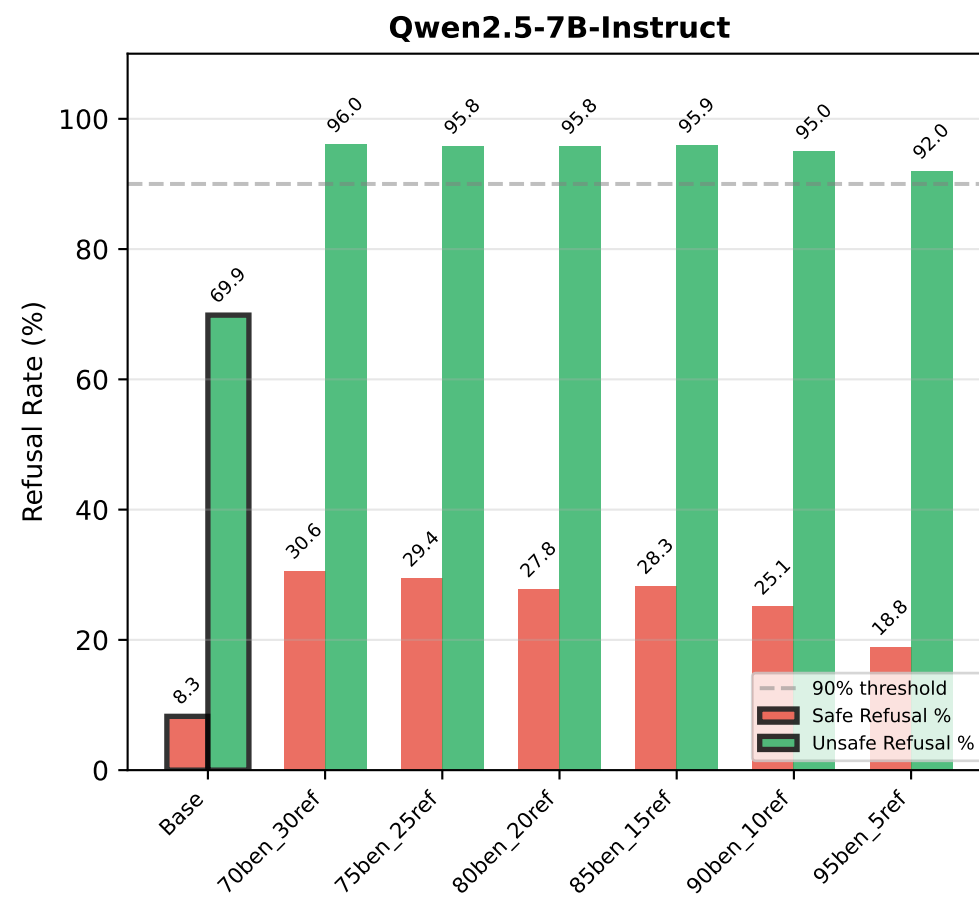
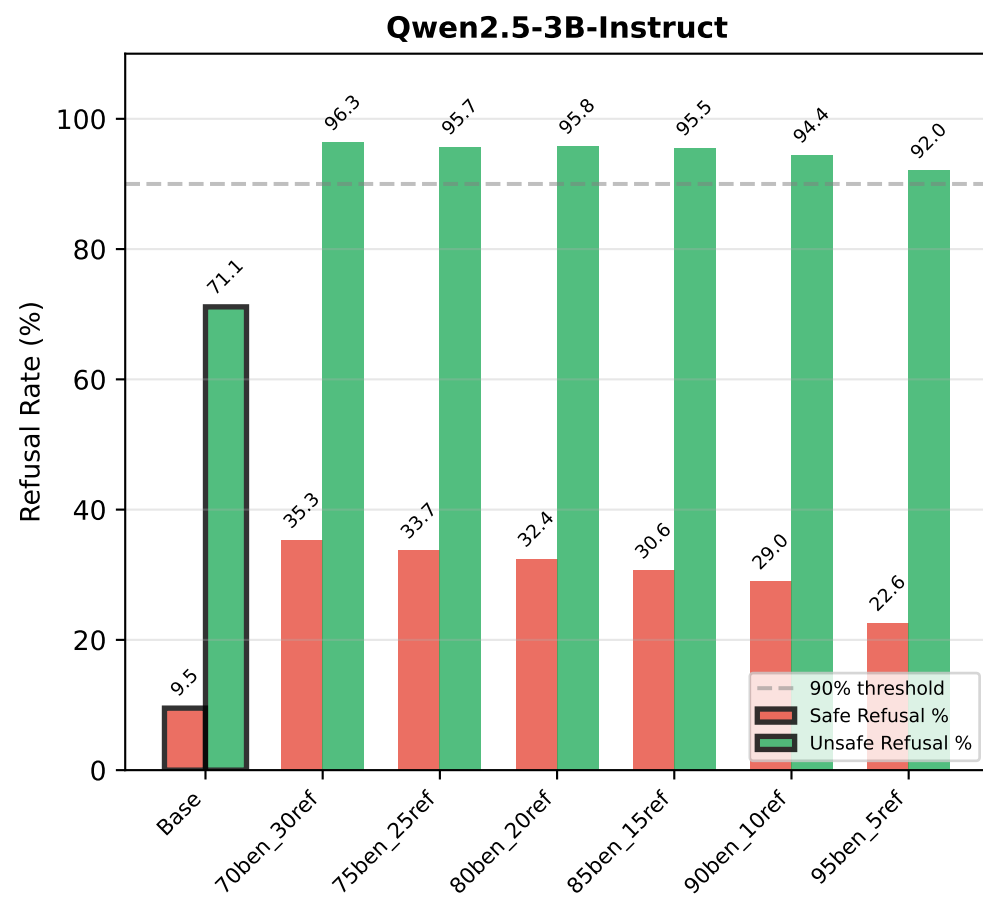
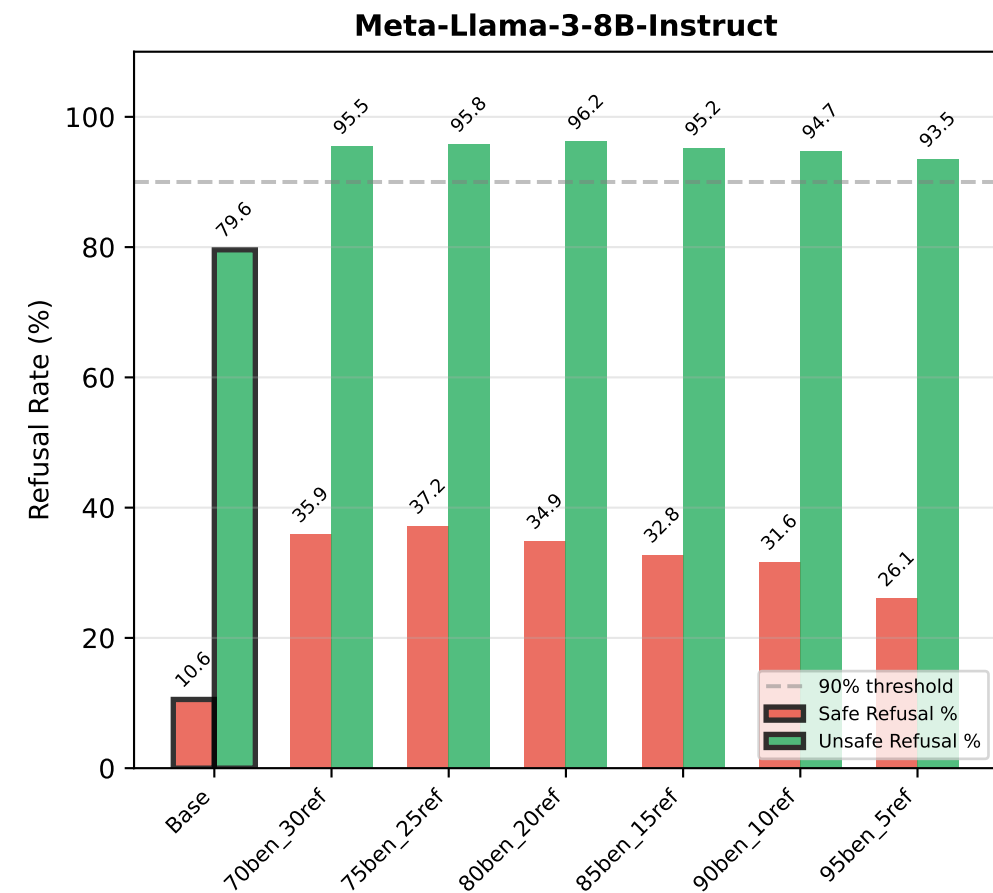
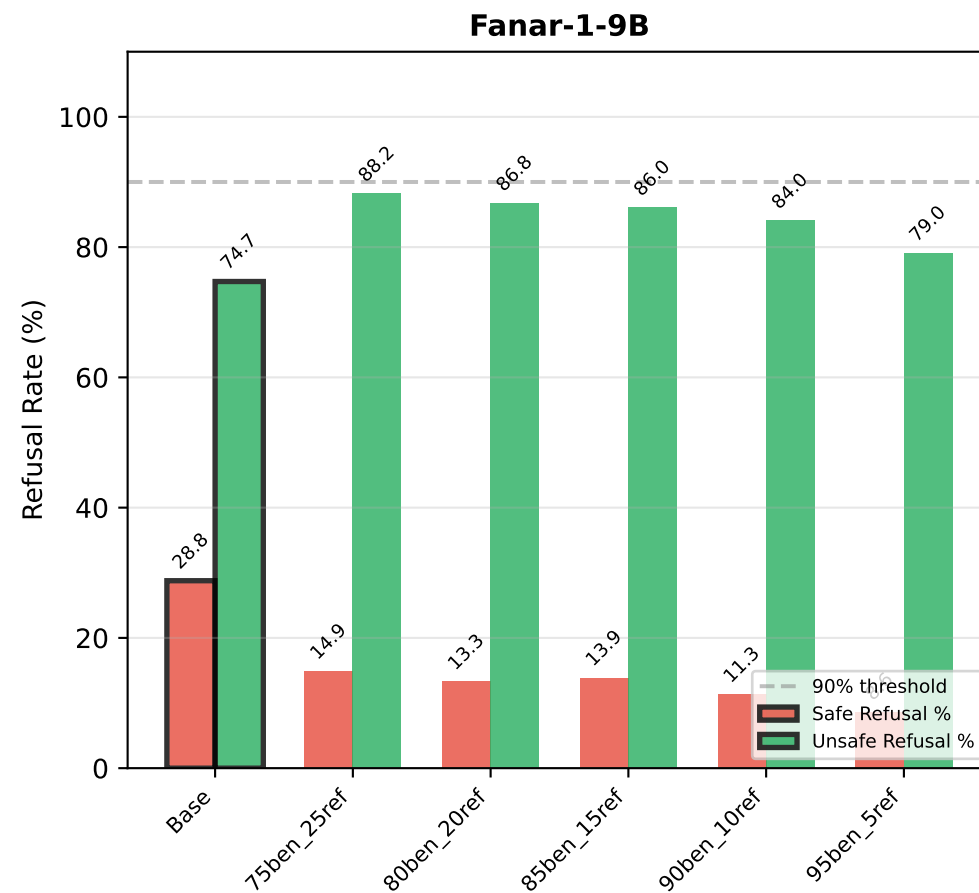
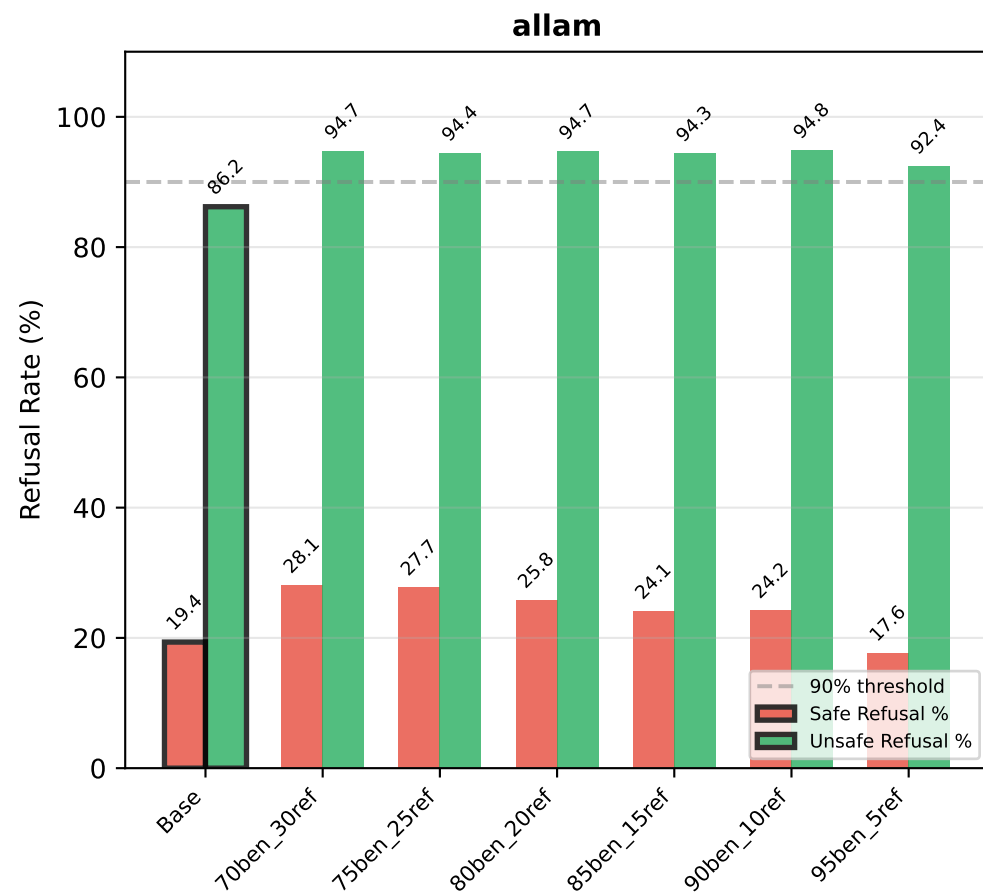


Curriculum Learning: Safe vs Unsafe Refusal Rates Across Training Stages



Key Insights:

- Safe Refusal (Red): Lower is better
Models should NOT refuse safe queries
- Unsafe Refusal (Green): Higher is better
Models SHOULD refuse unsafe queries
- Base: Original model performance
(highlighted with black border)
- Best stage for most models: 95ben_5ref
Lowest safe refusal while maintaining >90% unsafe refusal
- Fanar-1-9B: Best at 75ben_25ref
(highest unsafe refusal rate)